DISCRETE ELECTRONICS PROJECT

Audio Amplifier

PROJECT DEVELOPMENT, ORGANIZATION AND VERIFICATION PLAN

TEC Weeks 9–15

Duration:	7 weeks
Start:	TEC Week 9
End:	TEC Week 15
Team:	5 members
Organization:	1 Leader, 3 Developers and 1 Verifier
Main technology:	Discrete MOSFET transistors
Required quantity:	5 MOSFETs
Construction type:	Breadboard

Electronic Devices Course

Costa Rica Institute of Technology

Contents

1 General Project Description **2**

2 Objective of the Work Plan **2**

3 Team Organization **2**

4 Team Working Rules **3**

5 Detailed Schedule **4**

5.1 Project Week 1 – TEC Week 9 4

5.2 Project Week 2 – TEC Week 10 5

5.3 Project Week 3 – TEC Week 11 6

5.4 Project Week 4 – TEC Week 12 7

5.5 Project Week 5 – TEC Week 13 8

5.6 Project Week 6 – TEC Week 14 9

5.7 Project Week 7 – TEC Week 15 10

6 Preliminary Component List and Budget **11**

7 Five-MOSFET Control **11**

8 Main Milestones **12**

1 General Project Description

Based on the instructions received for the project, the following requirements will be considered from the beginning:

- A +12 V and −12 V power supply.
- A loudspeaker with the characteristics required for the project.
- An amplification stage developed using discrete devices.
- A total of **5 MOSFETs**.
- Complete construction of the prototype using a **breadboard**.
- Audio signal reproduction.
- Experimental verification of operation.
- Audibility tests.
- Documentation of the components used.

The specific function of each of the five MOSFETs is not arbitrarily established in this document.

During the first design stages, the final topology must be determined and the function performed by each device must be technically justified.

2 Objective of the Work Plan

The planning process follows this sequence:

Requirements → Design → Simulation → Construction → Integration → Testing → Delivery

The internal goal is to obtain a functional amplifier no later than project Week 5 (**TEC Week 13**).

Weeks 6 and 7 will be mainly used for final validation, corrections, documentation and preparation for the presentation.

3 Team Organization

lightgray **Team Member**	**Main Responsibility**
Leader	Planning, task distribution, weekly progress monitoring, purchase coordination, review of results, team integration, and communication with the client.
Developer 1	Audio input, volume control, signal conditioning, and compatibility with the audio source.
Developer 2	Preamplification, voltage amplification, driver stage, biasing, and analysis of the corresponding MOSFETs.
Developer 3	Power stage, ±12 V power supply, loudspeaker connection, output current, power dissipation, and thermal behavior.
Verifier	Test plan, independent measurements, comparison between simulation and measurements, datasheet review, and verification of project requirements.

4 Team Working Rules

To maintain constant progress and avoid delays during the development of the project, the following working rules are established:

1. Each team member will be responsible for completing the tasks assigned for each week.
2. If a team member encounters a problem that prevents progress, it must be communicated to the team so that a solution can be found before the next meeting.
3. Each developer must test their part of the circuit before connecting it to the other stages.
4. The main objective will be to achieve functional progress of the circuit every week, avoiding unnecessary documentation that does not directly contribute to the operation of the prototype.
5. The verifier will check the operation of the stages and perform the necessary measurements when required.
6. The leader will review the overall progress of the project before each meeting with the **client** and organize the tasks for the following week.
7. Problems found during construction must first be solved in the stage where they originate before modifying other parts of the circuit.
8. A stage with significant operating problems will not be integrated.
9. Important modifications made to the circuit must be communicated to the rest of the team to avoid working with different versions of the design.
10. Whenever possible, each meeting with the **client** should demonstrate functional progress compared with the previous week.

5 Detailed Schedule

5.1 Project Week 1 – TEC Week 9

Stage: Definition, research and requirements

Weekly objective:

Define exactly what must be built, which requirements must be met, and how the amplifier will be technically divided.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Review the complete project statement. Create the official list of requirements. Assign responsibilities. Create the schedule. Identify questions that must be discussed with the **client**. Organize the shared project folder.	Requirements list, schedule and distribution of responsibilities.
Developer 1	Research the audio source to be used, approximate input level, physical connection, volume control and necessary signal conditioning.	Input proposal and preliminary diagram.
Developer 2	Research amplification topologies using MOSFETs. Propose the preamplifier, voltage amplification and driver structure. Analyze how some of the five MOSFETs could be distributed.	Topology proposal and possible MOSFET functions.
Developer 3	Research the power stage, operation with ±12 V, loudspeaker, required current and thermal dissipation. Review practical limitations of building the power stage on a breadboard.	Preliminary analysis of power, current and construction.
Verifier	Transform the project requirements into measurable tests. Prepare a preliminary list of voltage, gain, power, frequency, clipping, temperature and audibility tests.	Preliminary test plan.

Weekly Deliverables

- Final list of known requirements.
- Questions to discuss with the client.
- Block diagram.
- Preliminary architecture.
- Distribution of responsibilities.
- MOSFET research.
- Preliminary component list.
- Preliminary test plan.

Result to Show During the Meeting

The team should be able to clearly explain:

$$\text{Input} \rightarrow \text{Amplification} \rightarrow \text{Driver} \rightarrow \text{Power} \rightarrow \text{Loudspeaker}$$

and explain how the design process will continue.

5.2 Project Week 2 – TEC Week 10

Stage: Design and simulation

Weekly objective:

Design the different stages and verify through simulation that they can operate correctly before physical construction.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Review the progress of the three stages. Confirm that the interfaces between blocks are compatible. Coordinate the component list and purchases. Verify that the five-MOSFET requirement is maintained.	Reviewed general schematic and consolidated component list.
Developer 1	Design and simulate the input, volume control and signal conditioning. Measure input and output amplitudes and verify that the signal is not distorted.	Simulation, V_{in}, V_{out} and gain values.
Developer 2	Design and simulate the preamplifier, voltage stage and driver. Determine operating points and the function of the MOSFETs assigned to these stages.	Schematic, simulations and operating points.
Developer 3	Design and simulate the power stage. Review output current, power supply, loudspeaker connection and expected power dissipation.	Power simulation and electrical/thermal estimates.
Verifier	Independently review the simulations. Compare the results with the requirements and create a table of theoretical values that will later be compared with the prototype.	Expected values table and observations.

Specific Review of the Five MOSFETs

By the end of this week, a table similar to the following should exist:

lightgray **MOSFET**	**Model**	**Stage**	**Function**
Q1			
Q2			
Q3			
Q4			
Q5			

A MOSFET should not be selected only to complete the required quantity. Each device must perform a technically justifiable function within the circuit.

Weekly Deliverables

- Functional simulation of each stage.
- Preliminary complete schematic.
- Five-MOSFET table.
- Component list.
- Expected theoretical values and main datasheets.

5.3 Project Week 3 – TEC Week 11

Stage: Modular breadboard construction

Weekly objective:

Physically build each block and verify its individual operation before integrating the complete amplifier.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Coordinate components and construction. Physically review the organization of the breadboards. Ensure that each stage is tested before integration. Record problems encountered.	Progress list and problem log.
Developer 1	Build the input and volume control on a breadboard. Apply a known signal and verify that the output is clean.	Photo of the construction and input/output measurements.
Developer 2	Build the preamplifier and driver. Measure biasing and gain. Compare the results with the simulation.	Working circuit and simulation–measurement comparison.
Developer 3	Build the power stage on a breadboard. Carefully review power connections, current and temperature. Begin testing with small signals.	Circuit, current measurements and temperature measurements.
Verifier	Perform independent measurements of DC voltages, amplitude, gain, noise and temperature. Record differences compared with the simulation.	Independent measurement table.

Completion Criterion

A stage can only be marked as:

READY FOR INTEGRATION

when it operates independently and its measurements are reasonable compared with the simulation.

5.4 Project Week 4 – TEC Week 12

Stage: Amplifier integration

Weekly objective:

Progressively connect all stages and obtain the first audio reproduction from the complete amplifier.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Direct the integration order. Create a list of faults. Assign each problem to a responsible team member and ensure that corrections are verified.	List of faults, responsible members and status.
Developer 1	Integrate the input with the amplifier. Correct level, noise or saturation problems coming from the source.	Integrated input and measured signal.
Developer 2	Integrate the preamplifier and driver. Correct biasing, gain, premature clipping or distortion problems.	Integrated intermediate stages operating correctly.
Developer 3	Integrate the power stage, power supply and loudspeaker. Monitor current and temperature during all tests.	Audio output and preliminary power measurements.
Verifier	Measure the complete system. Record gain, supply voltages, noise, observable distortion and temperature.	Integration report.

Recommended Order

$$\text{Input} \rightarrow \text{Preamplifier} \rightarrow \text{Driver} \rightarrow \text{Power} \rightarrow \text{Loudspeaker}$$

Testing should not begin immediately at maximum power.

Deliverable

First complete demonstration of audio reproduction.

5.5 Project Week 5 – TEC Week 13

Stage: Testing and corrections

Weekly objective:

Quantitatively verify the operation of the amplifier and correct any requirements that are not yet satisfied.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Compare results with requirements. Prioritize problems. Assign corrections and ensure that every modification is tested again.	Updated compliance matrix.
Developer 1	Correct input noise, volume control, signal level and compatibility with the audio source.	Measurements before and after corrections.
Developer 2	Correct gain, frequency response, biasing and distortion problems. Complete the technical justification of the MOSFETs used.	Corrected results and MOSFET justification.
Developer 3	Correct power, clipping, heating, power supply and loudspeaker output problems.	Power, clipping and temperature measurements.
Verifier	Execute the complete testing procedure and independently record all results.	Official results table.

Minimum Tests

1. Power supply.
2. Biasing.
3. Gain.
4. Output power.
5. Clipping.
6. Frequency response.
7. Noise.
8. Temperature.
9. Audio reproduction.
10. Audibility.

For frequency characterization, the following frequencies may initially be considered:

$$100, \quad 250, \quad 500, \quad 1000, \quad 2000, \quad 5000, \quad 10000 \text{ Hz}$$

adjusting the test points according to the documented system range.

Internal Goal

By the end of this week, the amplifier should be:

FUNCTIONAL

although minor adjustments may still be required.

5.6 Project Week 6 – TEC Week 14

Stage: Final validation

Weekly objective:

Obtain repeatable results and resolve pending faults without making unnecessary last-minute modifications.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Review every requirement, resolve pending items, organize documentation and verify that evidence exists for the characteristics that will be presented.	Final requirements list and documentation.
Developer 1	Perform final adjustments to the input, volume, wiring and signal quality.	Final validated input.
Developer 2	Perform final amplification adjustments and complete calculations and justifications for the five MOSFETs.	Final technical documentation.
Developer 3	Review the reliability of the breadboard construction, power supply, power connections and temperature.	Final physical and thermal review.
Verifier	Independently repeat all critical tests and classify each requirement.	Final verification report.

Acceptance Matrix

lightgray **Requirement**	**Compliant**	**Pending**	**Not Compliant**
±12 V power supply			
Five MOSFETs used and justified			
Gain			
Power			
Frequency response			
Audibility			
Temperature			
Audio quality			
Stable breadboard construction			

5.7 Project Week 7 – TEC Week 15

Stage: Closing and presentation

Weekly objective:

Present a previously tested amplifier with organized documentation and a prepared demonstration.

lightgray **Responsible**	**Assigned Work**	**Expected Evidence**
Leader	Prepare the presentation, distribute presentation sections, coordinate the rehearsal, organize the demonstration and prepare possible answers for the **client**.	Complete presentation and completed rehearsal.
Developer 1	Explain the audio source, input, signal conditioning and volume control. Support the demonstration.	Prepared presentation section.
Developer 2	Explain the operation of the amplification stages and the function of the MOSFETs used.	Prepared technical explanation.
Developer 3	Explain the power supply, power stage, loudspeaker, current and thermal behavior.	Prepared power-stage explanation.
Verifier	Present the final measurements and evidence of compliance. Perform a final operational check before the presentation.	Final report and final verification.

Final Deliverables

- Completed amplifier on breadboard.
- Five MOSFETs installed and justified.
- Audio demonstration.
- Experimental measurements.
- Datasheets.
- Technical documentation.
- Presentation.

6 Preliminary Component List and Budget

lightgray **Component**	**Qty.**	**Unit Price**	**Subtotal**	**Intended Use**
MOSFET	5	TBD	TBD	Amplification and power stage according to final design.
Large breadboard	2–3	≈ ¢5 900	≈ ¢11 800–17 700	Complete prototype construction.
Resistors	Various	–	≈ ¢1 500–3 000	Biasing, feedback and gain adjustment.
10 µF capacitor	4	≈ ¢89	≈ ¢356	Coupling and filtering.
100 µF capacitor	4	≈ ¢199	≈ ¢796	Filtering and decoupling.
470 µF capacitor	2	≈ ¢295	≈ ¢590	Filtering.
2200 µF capacitor	2	≈ ¢520	≈ ¢1 040	Power-supply filtering.
0.1 µF capacitor	6	≈ ¢110	≈ ¢660	Decoupling and stability.
Potentiometer	1–2	TBD	≈ ¢500–1 500	Volume control and possible adjustments.
Heat sinks	As required	≈ ¢420	TBD	Temperature control for MOSFETs that require them.
Breadboard jumper wires	1 set	TBD	≈ ¢1 500–3 000	Prototype interconnection.
Audio connectors	As required	TBD	TBD	Audio-source connection.
Loudspeaker	1	TBD	TBD	Amplifier output load.
+12 V / −12 V supply	1	TBD	TBD	Amplifier power supply.

Note: The cost of the five MOSFETs, loudspeaker and power supply is not yet included in a general total because the final models must be established after the design has been validated.

7 Five-MOSFET Control

Because the project must use five MOSFETs, the following control table will be maintained throughout development:

lightgray **ID**	**Model**	**Stage**	**Justification**
Q1			
Q2			
Q3			
Q4			
Q5			

8 Main Milestones

lightgray **Project Week**	**TEC Week**	**Milestone**
1	9	Requirements and architecture defined.
2	10	Complete design simulated and five MOSFETs defined.
3	11	Modules built and tested on breadboard.
4	12	First complete amplifier operating.
5	13	Complete testing and functional prototype.
6	14	Final validation and documentation.
7	15	Presentation and final delivery.